Acute Change in Renal Markers

CASE 04 | FICTIONAL PERSON | NO REAL PATIENT DATA

Synthetic model-evaluation document. Not for clinical use. Do not use these values to make decisions about a real person.

Emergency department note

A 76-year-old man presents after two days of repeated vomiting, loose stools, and markedly reduced oral intake. He reports lightheadedness when standing. There is no fever, dysuria, flank pain, or visible blood in urine.

Examination note: dry oral mucosa and orthostatic pulse increase. No edema, pulmonary crackles, or bladder distension is recorded.

Kidney history	No known chronic kidney disease. A serum creatinine of 0.88 mg/dL was recorded one month earlier.
Recent exposures	No recent contrast study, NSAID use, new prescription, or urinary catheter.
Body mass	70 kg at the current visit.
Urine output	160 mL over the first 8 hours of observation; no anuria.
Imaging	Renal ultrasound: no hydronephrosis; bladder not distended.

All names, dates, histories, values, intervals, and notes in this case are fabricated.

Renal chemistry trend

The current sample was collected approximately 48 hours after the onset of gastrointestinal symptoms. Prior baseline was obtained at the same laboratory.

Test	Result	Unit	Printed reference interval	Report flag
Creatinine, baseline	0.88	mg/dL	0.70 - 1.30	
Creatinine, current	1.34	mg/dL	0.70 - 1.30	H
BUN, current	34	mg/dL	7 - 20	H
Sodium, current	146	mmol/L	135 - 145	H
Potassium, current	4.9	mmol/L	3.5 - 5.1	
Bicarbonate, current	20	mmol/L	22 - 29	L
eGFR, baseline	89	mL/min/1.73 m2	>= 60	

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Urine findings and time course

Urine output averaged 0.29 mL/kg/hour over eight hours (160 mL divided by 70 kg and 8 hours). The record contains no prior oliguria.

Test	Result	Unit	Printed reference interval	Report flag
Urine specific gravity	1.030		1.005 - 1.030	
Urine protein	Negative		Negative	
Urine blood	Negative		Negative	
Urine RBC	0 - 1	/hpf	0 - 2	
Urine casts	0 - 2 hyaline	/lpf	0 - 2 hyaline	
Urine ketones	Trace		Negative	
Repeat creatinine	Not yet available			

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